

Jay H. Jivandas

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EDUCATION

The Ohio State University, College of Engineering (3.6 GPA)	Columbus, OH
<i>Bachelor of Science in Computer Science and Engineering, AI Specialization</i>	<i>May 2026</i>
<i>M.S. Computer Science and Engineering (BS/MS Program)</i>	<i>May 2027</i>
<i>Minors: Statistics and Business; Engineering Honors</i>	
<i>Honors Research Distinction; Dean's List (5/7 semesters)</i>	

PROFESSIONAL EXPERIENCE

Imageomics Institute	Columbus, OH
<i>Software Engineering Intern</i>	<i>Oct 2025 – Present</i>
- Contributing to the distributed-downloader and Tree of Life Toolbox to support large-scale biodiversity dataset acquisition for ML research - Improving pipeline portability across HPC systems by standardizing dependencies and reusable workflows	
SpiceBox.AI	Dubai, UAE
<i>AI/ML Development Intern</i>	<i>May 2025 – Present</i>
- Building foundational components for an AI-powered law firm, aligning with the UAE's vision for an AI-driven city - Developed tools to automate legal workflows for SMEs, reducing overhead by over 50%	
Lifestyle Communities	Columbus, OH
<i>Application Development Intern</i>	<i>May 2024 – July 2024</i>
- Created reusable Laravel/PHP/JS components, saving 15+ dev hours per project - Rebuilt coupon management system with senior developers to streamline discount generation, distribution, and tracking	
BMB Group	Dubai, UAE
<i>Full Stack Web Development Intern</i>	<i>June 2023 – Aug 2023</i>
Rebuilt maintenance app (PHP/MySQL/JS) used by 1,000+ employees, improving repair efficiency by 20%	

RESEARCH EXPERIENCE

Imageomics Institute + Infectious Diseases Institute, The Ohio State University	Columbus, OH
<i>Undergraduate Research Assistant</i>	<i>May 2025 – Present</i>
- Leading day-to-day development of a tick species classification pipeline with ~15 researchers from Imageomics and IDI - Built a BioCLIP embedding pipeline (PyBioCLIP) and trained few-shot SVM classifiers (0–25 samples/class), improving accuracy 60% → 89% - Developed Python pipelines to process 2,000+ tick images and support ongoing ML experiments for species identification	
AgSensing Data Analytics for Agriculture, The Ohio State University	Columbus, OH
<i>Undergraduate Research Assistant</i>	<i>Jan 2025 – May 2025</i>
Preparing large-scale agricultural datasets for ML models to predict optimal nitrogen rates	
Imageomics Institute, The Ohio State University	Columbus, OH
<i>Undergraduate Research Assistant</i>	<i>Aug 2024 – Dec 2024</i>
Evaluated deep learning and computer vision models (SAM, YOLO, TensorFlow, PyTorch) with PhD students on 1,000+ butterfly images, applying bounding boxes for species identification	
Civil, Environmental and Geodetic Engineering Department, The Ohio State University	Columbus, OH
<i>Undergraduate Research Assistant</i>	<i>Aug 2023 – April 2024</i>
Collected and processed raw flux data from two ecosystem measurement sites using MATLAB	

PROJECTS

S&P 500 Market Prediction Model
- Built a machine learning model to forecast S&P 500 trends using Yahoo Finance data and engineered market indicators - Achieved ~60% accuracy through iterative tuning, feature selection, and historical backtesting
OSU Engineering Org Assistant
Built a chat-based assistant using Llama API to help students find engineering orgs based on interests and queries

QUALIFICATIONS

- Programming Languages:** Python, Java, C, C++, SQL, PHP, JavaScript, HTML/CSS, MATLAB, R, Ruby, Bash
- Frameworks & Libraries:** PyTorch, TensorFlow, scikit-learn, pandas, NumPy, PIL, YOLO, SAM, React Native, Hugging Face, PyBioCLIP
- Tools & Platforms:** Git, Linux, Tableau, SolidWorks
- Relevant Coursework:** Machine Learning, Computer Vision, Statistical Modeling for Discovery I (STAT 3301), Data Structures and Algorithms, Software Development, Database Systems, Intro to AI, Linear Algebra, Calculus I & II, Discrete Structures, Foundations of Higher Mathematics

LEADERSHIP & ACTIVITIES

OSU Boxing Club, Student Athlete	<i>Aug 2025 – Present</i>
OSU Undergraduate Honors Committee, Student member (1 of 6)	<i>Aug 2023 – May 2025</i>
FIRST Alumni and Robotics, Student Mentor for a group of high school students	<i>Oct 2023 – April 2024</i>